

LMUTH SATS 24-HOUR OUTCOMES

Research Data Collection Platform

PHASE 1 — DESIGN & ARCHITECTURE APPROVAL DOCUMENT

Study: Association Between South African Triage Scale (SATS) Categories and 24-Hour Hospital Outcomes at Levy Mwanawasa University Teaching Hospital Emergency Department

Study period: 1 July 2025 – 30 June 2026 | Expected scale: approximately 8,000 retrospective ED records

Item	Proposed decision
Mobile application	Flutter — Android, offline-first
Backend / API	Node.js — secure REST/JSON API
Central database	MongoDB — structured research repository
Administration	React (ReactJS) — responsive web dashboard for PI / QC / Admin
Operating model	Multi-user, offline-first, incremental central synchronisation

Purpose of this document

This document presents the proposed Phase 1 architecture, data model, screen flow, wireframe specification, offline/synchronisation design, security model, central repository, export structure and technical questions for research-team approval before full MVP development begins.

*Confidential — Research System Specification
Prepared for Research Team / Principal Investigator approval*

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0. Plain-Language Summary (for non-technical readers)

This section explains the proposal in everyday terms. Every technical detail below is explained in full in the later sections for the record, but nothing here requires a technical background to follow.

We are proposing to build a phone app for four (or more) Research Assistants (RAs) to enter clinical data from about 8,000 old Emergency Department records into a research database, so it can later be analysed.

- ✓ The app works even with no internet — an RA can keep entering data anywhere in the hospital, and the app safely stores it on the phone until it can connect and send the data to a central, secure database.
- ✓ Every entry is linked to the RA who made it, the phone/device it was made on, and the exact time it was made — so nothing is ever anonymous or untraceable inside the system.
- ✓ A separate Quality Control (QC) reviewer independently re-checks at least 1 in 10 records to catch mistakes, without seeing the original RA's answers first, so the check is fair.
- ✓ Patient-identifying information (like a hospital file number) is stored in a locked-away, separate part of the system and is never included in the data that goes to the researchers for analysis.
- ✓ At the end, the Principal Investigator (PI) can download a clean spreadsheet (CSV/Excel) ready for statistical analysis, plus a 'codebook' that explains what every column means.
- ✓ This document is Step 1 only: a plan for the research team to review and approve. No app code will be written until this plan is signed off.

1. Executive Summary

The LMUTH SATS Research Data Platform is proposed as an analysis-oriented research abstraction system for approximately 8,000 retrospective Emergency Department records. It will support at least four Research Assistants working independently on Android devices, including periods with no internet connectivity, and will synchronise completed and pending records securely to one authoritative central repository.

The design follows the non-negotiable rule set out in the Developer Brief: the application is not designed by digitising the photographed clinical SATS sheet. The approved research Data Abstraction Tool, the research questions, the data dictionary and the analytical model are the primary design inputs. The clinical SATS/TEWS sheet is treated only as the source record an RA reads from.

This Phase 1 submission exists to obtain research-team agreement on the data model and workflow before full MVP coding begins, as the Developer Brief specifically requires approval of the data dictionary and screen flow first.

The proposed technology stack is Flutter + Node.js + MongoDB. The source brief itself recommends PostgreSQL but permits an equivalent technology where functional requirements are still met. The MongoDB design below therefore uses explicit schemas, server-side validation, indexed identifiers, audit records, version control, controlled relationships between collections, and analysis-ready exports, so that the same guarantees a relational database would give are preserved.

2. Design Principles & Requirements

- ✓ Offline-first: Data entry must continue without internet; connectivity loss must never cause data loss or force re-entry.
- ✓ Central synchronisation: Only incrementally created/changed records, assignments, QC updates and configuration changes are synchronised — not the whole dataset every time.
- ✓ Auditability: Every record and every change is attributable to a user, device, timestamp and application version.
- ✓ Privacy: Direct identifiers and raw linkage identifiers are excluded from analytical exports and are access-restricted.
- ✓ Quality control: At least 10% independent QC, discrepancy comparison and resolution, verification and locking.
- ✓ Analysis readiness: PI-authorized exports provide clean, de-identified CSV/XLSX datasets and a machine-readable codebook.
- ✓ Scale: Architecture targets at least 100,000 records and supports additional Research Assistants beyond the initial four.

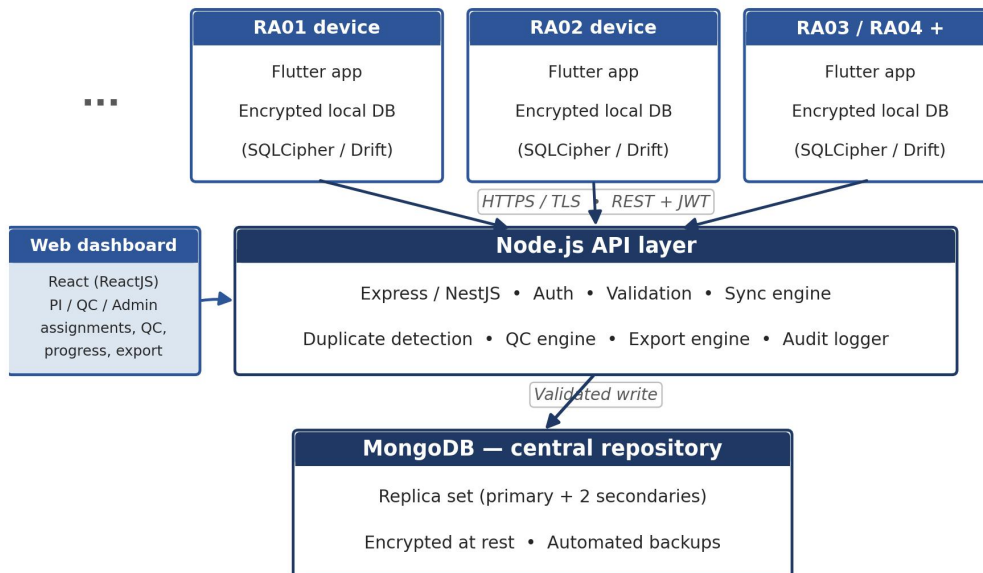
3. Proposed Solution Architecture

3.1 Logical architecture

Layer	Technology / responsibility
Mobile client	Flutter Android app: offline-first forms, local encrypted database, assignment/work queues, validation, sync manager, device registration.

Layer	Technology / responsibility
Local persistence	Encrypted SQLite (SQLCipher via Drift). Stores drafts, completed records awaiting sync, the sync queue, and relevant configuration.
API layer	Node.js secure REST/JSON API over HTTPS/TLS. Handles authentication, authorisation, server-side validation, synchronisation, duplicate checks, QC and export requests.
Central repository	MongoDB. Structured collections, validation rules, indexes, version fields, audit and sync logs, backups and controlled access.
Web administration	React (ReactJS) responsive PI/QC/Admin dashboard for assignments, progress, missingness, QC, duplicate review, user/device management and export.
Export layer	CSV/XLSX master data, codebook, QC report, missing-data report, exclusion log, fieldwork progress report; optional SPSS .sav if approved.

3.2 Data flow



Boundary: the Flutter client and the web dashboard never connect directly to MongoDB — every request passes through the authenticated Node.js API.

Important boundary: the Flutter client never connects directly to MongoDB. All access to the central database happens only through the authenticated Node.js API.

4. MongoDB Database Architecture

The central repository uses separate collections for core research data, restricted linkage information, operational management, QC and auditability, so the explicit structure the analytical model needs is preserved even though MongoDB stores documents rather than rows.

Collection	Purpose
users	Individual accounts, roles and account status.
devices	Registered device identifiers, authorisation and deactivation state.
assignments	Research Assistant batch allocations, date/page/file ranges and workload.
research_records	Core one-record-per-eligible-presentation research abstraction record.
patient_linkage	Restricted mapping between Study ID and hospital-record identifier; never included in analytical exports.
outcomes	24-hour outcome evidence, source and verification information.
qc_reviews	Independent QC abstraction, comparisons, discrepancies and resolution.
audit_logs	Field-level change history with previous/new values and reason.
sync_logs	Device synchronisation attempts, results and errors.
app_versions	Application and data-dictionary versions and release information.
exclusions	Screened-out records with an exclusion reason code.

Identity strategy: each record receives a human-readable Study ID such as LMUTH-SATS-RA01-000001 and an internal globally unique ID (UUID). A simple local sequence number is never used on its own, because multiple devices may create records at the same time while offline.

5. Proposed Core Data Schema

The table below is the full analytical core, confirmed against the approved Data Abstraction Tool (the study's structured abstraction instrument) and following its own canonical variable order for the one-row-per-presentation analysis dataset.

Variable	Type	Coding / format	Research role
study_id	Text	Unique generated Study ID	Primary key
ed_date	Date	DD/MM/YYYY	Descriptive
ed_time	Time	HH:MM	Process/context
triage_time	Time	HH:MM / missing	Process
age	Numeric	Completed years	Covariate
sex	Categorical	1 Male; 2 Female; 9 Not recorded	Covariate
referral	Categorical	0 Direct; 1 Referred; 9 Unknown	Covariate
dm	Categorical	0 No; 1 Yes; 9 Unknown	Covariate component

Variable	Type	Coding / format	Research role
htn	Categorical	0 No; 1 Yes; 9 Unknown	Covariate component
asthma	Categorical	0 No; 1 Yes; 9 Unknown	Covariate component
rvd	Categorical	0 No; 1 Yes; 9 Unknown	Covariate component
other_comorb	Categorical	0 No; 1 Yes; 9 Unknown	Covariate component
other_comorb_text	Text	Free text (if other_comorb = 1)	Descriptive
comorb_any	Binary	0 No; 1 Yes; 9 Unknown	Covariate (derived)
preg_test	Categorical	0 Negative; 1 Positive; 8 Not done/N-A; 9 Not recorded	Context, where applicable
temp	Numeric	Degrees C	Secondary clinical
hr	Numeric	beats/min	Secondary clinical
rr	Numeric	breaths/min	Secondary clinical
sbp	Numeric	mmHg	Secondary clinical
dbp	Numeric	mmHg	Secondary clinical
spo2	Numeric	0–100%	Secondary clinical
rbs	Numeric	mmol/L	Secondary clinical
rdt	Categorical	0 Negative; 1 Positive; 8 Not done; 9 Not recorded	Secondary clinical
mobility	Categorical	0 Walking; 1 With help; 2 Stretcher/immobile; 9 Missing	Secondary clinical
avpu	Categorical	0 Alert; 1 Voice; 2 Pain; 3 Unconscious; 9 Missing	Secondary clinical
trauma	Categorical	0 No; 1 Yes; 9 Unknown	Secondary clinical
tews_total	Numeric	Recorded score	Secondary

Variable	Type	Coding / format	Research role
			clinical
complaint_text	Text	Verbatim chief complaint	Descriptive (see Technical Question)
complaint_group	Categorical	Frozen 1–10 category list — see §5.3	Secondary clinical
multiple_complaint	Binary	0 No; 1 Yes	Descriptive
discriminator_yes	Binary	0 No; 1 Yes	Secondary clinical
discriminator_type	Categorical	Frozen 0–18 code list — see §5.3	Secondary clinical
sats_cat	Categorical	1 Green; 2 Yellow; 3 Orange; 4 Red	Primary IV
sats_complete	Binary	0 No; 1 Yes	Quality
initial_destination	Categorical	Approved destination codes	Context only
clinician_time	Time	HH:MM, only if documented	Process
treatment_time	Time	HH:MM, only if documented	Process
outcome24	Categorical	1 ED discharge; 2 Ward; 3 HDU/ICU; 4 Death	Primary DV
outcome_datetime	Date/Time	DD/MM/YYYY HH:MM	Process
outcome_source	Categorical	Frozen 1–6 list — see §5.3	Verification
outcome_verified	Binary	0 No; 1 Yes	Quality
miss_sats	Binary	0 No; 1 Yes	Quality
miss_tews	Binary	0 No; 1 Yes	Quality
miss_vitals	Binary	0 No; 1 Yes	Quality
miss_outcome	Binary	0 No; 1 Yes	Quality
source_conflict	Binary	0 No; 1 Yes	Quality
qc_verified	Binary	0 No; 1 Yes	Quality
qc_reviewer	Text	Reviewer ID	Audit
extractor_id	Text	Auto-captured user ID	Audit
abstract_date	Date	Auto-captured	Audit
device_id	Text	Auto-captured device ID	Audit

Variable	Type	Coding / format	Research role
app_version	Text	Auto-captured version	Audit

Missing-value rule: explicit missing codes (e.g. 9 = unknown/not recorded) are used only for categorical fields as shown above. Continuous measurements (temp, hr, rr, sbp, dbp, spo2, rbs) are left system-missing/blank with the corresponding miss_ flag set — zero is never entered unless zero is the true clinical value.*

5.1 Full MongoDB document schema — research_records

The nested document below is how the flat variable table above is actually stored and validated in MongoDB. Sections are grouped because they are always read/written together as one abstraction unit; outcomes, qc_reviews and audit_logs are kept as separate collections referencing this record (see §4), because they are created at different times by different roles and must never be edited by the Research Assistant after lock.

```
{
  _id: ObjectId,
  study_id: "LMUTH-SATS-RA01-000001", // human-readable, generated server-side on first sync
  linkage_id: ObjectId, // ref -> patient_linkage, NEVER included in export
  status: "Draft" | "Clinical Data Complete - Outcome Pending" | "Complete" |
    "Pending Sync" | "Sync Failed" | "Synced" | "QC Required" |
    "Returned for Correction" | "Verified" | "Locked" | "Excluded",
  mode: "PILOT" | "TRAINING" | "PRODUCTION",
  extractor_id: String, abstract_date: Date, device_id: String, app_version: String,
  data_dictionary_version: String, version: Number, // optimistic-concurrency counter
  created_at, updated_at: Date,

  eligibility: {
    ed_date: Date, ed_time: String, triage_time: String, age: Number, eligible: Boolean,
    exclusion_code: "1"|"2"|"3"|"4"|"5"|"9", exclusion_reason: String // see §5.3
  },
  patient: {
    sex: "1"|"2"|"9", referral: "0"|"1"|"9",
    comorbidities: { dm, htn, asthma, rvd, other: "0"|"1"|"9", other_text: String },
    comorb_any: "0"|"1"|"9", preg_test: "0"|"1"|"8"|"9"
  },
  sats: {
    sats_cat: "1"|"2"|"3"|"4", tews_total: Number, discriminator_yes: Boolean,
    discriminator_type: String, // code 0-18, see §5.3
    documentation_complete: Boolean // sats_complete
  },
  physiology: {
    temp, hr, rr, sbp, dbp, spo2, rbs: Number, rdt: "0"|"1"|"8"|"9",
    mobility: "0"|"1"|"2"|"9", avpu: "0"|"1"|"2"|"3"|"9", trauma: "0"|"1"|"9"
  },
  presentation: {
    chief_complaint_verbatim: String, complaint_group: String, // see §5.3
    multiple_complaints: Boolean
  }
}
```

```

},
initial_destination: String,          // Resus/Surgical/Medical/... — NOT the DV
process: {
  clinician_time: String, treatment_time: String // only where actually documented
},
data_quality: {
  miss_sats: Boolean, miss_tews: Boolean, miss_vitals: Boolean, miss_outcome: Boolean,
  source_conflict: Boolean, qc_required: Boolean, reviewer_id: ObjectId, qc_comment: String
},
duplicate_flags: [ { matched_record_id: ObjectId, basis: [String], resolved: Boolean } ]
}

```

5.2 Preserving relational integrity in MongoDB

- Multi-document ACID transactions are used for any write that touches more than one collection (e.g. record lock + audit entry; sync write + sync_log entry) so a record is never marked “Synced” without a matching sync_log, or locked without an audit trail.
- All cross-collection references use MongoDB ObjectId references (not embedding), validated at the API layer — functionally equivalent to foreign keys.
- JSON Schema validation is attached to each collection at the database level (not only in application code), so sats_cat, outcome24, sex, etc. can only ever be written as one of the approved codes below.
- The analysis export (§14) is generated by an aggregation pipeline that joins research_records + outcomes into exactly one row per eligible ED presentation, without the raw operational documents ever being flattened for RA-facing use.

5.3 Frozen code lists (per the approved Data Abstraction Tool)

These lists are frozen per the study's approved abstraction instrument and are hard-coded into both server- and client-side validation (§11, §12) so a device can never submit a code outside these lists.

Primary complaint_group

Code	Group
1	Trauma/injury
2	Cardiovascular/chest pain
3	Respiratory
4	Neurological
5	Gastrointestinal/abdominal
6	Infectious/fever
7	Obstetric/gynaecological
8	Endocrine/metabolic
9	Poisoning/toxicological
10	Other

Clinical discriminator_type

Code	Discriminator group
0	No discriminator documented
1	Shock/uncontrolled haemorrhage
2	Chest pain/cardiovascular
3	Seizure/altered consciousness
4	Major trauma/fracture/dislocation
5	Penetrating injury
6	Burns
7	Poisoning/overdose
8	Hypoglycaemia
9	Hypertensive emergency
10	Respiratory distress/shortness of breath
11	Haemoptysis
12	Abdominal pain/trauma
13	Pregnancy-related emergency
14	Bradycardia
15	Controlled haemorrhage
16	Persistent vomiting
17	PV bleeding
18	Other documented discriminator

Primary outcome_source

Code	Source
1	ED record
2	Patient file
3	Ward register
4	HDU/ICU register
5	Mortality record
6	Multiple sources

6. Research Model Preservation

Research element	Operational variable(s)	Planned analysis
Primary independent variable	SATS category: Green, Yellow, Orange, Red	Primary exposure
Dependent variable	24-hour outcome: ED discharge, General Ward, HDU/ICU, Death	Primary outcome
Primary covariates	Age, sex, referral status, comorbidities	Adjusted model
Secondary/exploratory	TEWS, temperature, HR, RR, SBP/DBP, SpO2, RBS, mobility, AVPU, trauma, complaint group, discriminator	Separate clinical/physiological analyses
Primary inferential model	Outcome24 ~ SATS + age + sex + referral + comorbidity	Multinomial logistic regression

The immediate ED destination (e.g. Resus, Surgical, Medical, Ward) is treated only as context/service destination and is NOT the study's dependent variable. The 24-hour outcome must be established from linked downstream ED/ward/HDU/ICU/mortality records.

7. Eligibility & Outcome Decision Rules

7.1 Exclusion codes

Code	Reason
1	Age below 16
2	Dead on arrival / Blue
3	SATS category missing
4	24-hour outcome indeterminable
5	Transfer prevents outcome ascertainment
9	Other — specify (exclusion_reason free text)

7.2 24-hour outcome hierarchy

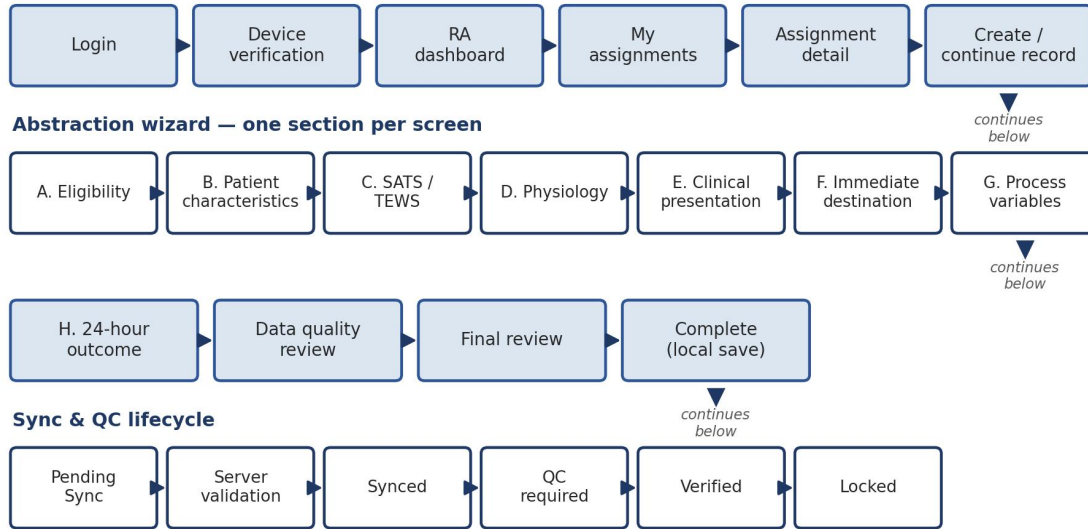
Death → HDU/ICU → General Ward → ED Discharge. If multiple events occur during the first 24 hours, the highest-priority outcome in this hierarchy is recorded. Example: ward admission followed by death at 6 hours = Death; ward admission followed by ICU transfer at 8 hours = HDU/ICU.

8. User Roles & Permission Matrix

Permission	RA	QC Reviewer	PI	Administrator
Individual login / device	✓	✓	✓	✓
Create/edit own drafts	✓	—	—	—
Complete and sync assigned records	✓	QC records	—	—
View own assignments	✓	—	✓	✓
Correct records returned by QC	✓	—	✓	—
Independent QC abstraction	—	✓	View only	—
Compare / resolve discrepancies	—	✓	✓	—
Verify / lock records	—	✓	✓	—
Reopen locked record with reason	—	—	✓	—
View all progress	—	Assigned only	✓	✓
Assign batches	—	—	✓	—
Master export	✗	✗	✓	Authorised only
Manage users/devices	✗	✗	Limited	✓

9. Mobile Screen Flow

The mobile workflow follows the research abstraction process rather than the physical layout of the photographed SATS sheet:



Alternative queues, reachable from the RA dashboard at any point: Drafts • Pending Outcome • Pending Sync • Sync Failed • Returned by QC

10. Wireframe Specification

The wireframe inventory below will be used for detailed UI approval and subsequent implementation. Full pixel-level mockups follow as a companion visual document once this structure is approved.

Area	Screens / purpose
Authentication	Login; device verification; session state; optional PIN/biometric re-entry.
RA dashboard	Assigned, completed, remaining, synced, pending sync, QC returned, today's/weekly activity.
Assignments	Assignment list, assignment detail, progress and workload.
Abstraction	Eligibility; identification; patient characteristics; SATS/TEWS; physiology; clinical presentation; destination; process variables; 24-hour outcome; data quality; review/submit.
Record management	Drafts; pending outcome; pending sync; sync errors; returned QC; record detail.
QC	Assigned QC records; blind re-abstraction; comparison; discrepancy resolution; verification/lock.
PI/Admin web (React)	Dashboard; assignments; users; devices; records; QC; duplicates; missingness; audit; exports; configuration.

UI principles: mobile-first layout, clear required/optional states, explicit Missing/Not Recorded values where appropriate, soft warnings for unusual-but-possible clinical values, hard validation for

impossible values, and a persistent local-save indicator so collectors always know their work is safe, even offline.

11. Offline Storage & Synchronisation Design

Local record states: Draft → Complete / Outcome Pending → Pending Sync → Syncing → Synced, with Sync Failed retained locally until a successful retry. QC/returned/locked states flow down from the central workflow.

- Every record is written to local encrypted storage before it is considered safely captured.
- Sync is incremental and sends only new/changed records, assignments, QC updates and configuration changes.
- Server acknowledgement is required before a record is marked Synced.
- An interrupted upload leaves the local record intact and retryable — nothing is lost.
- Sync requests are idempotent, so a retried request cannot create a duplicate record.
- Optimistic concurrency / version checks prevent one change silently overwriting another.
- Technical sync logs record user, device, record ID, timestamp, status and error detail.

11.1 Conflict handling

Situation	System response
New record from offline device	Create using UUID + Study ID; acknowledge only after validation.
Retry after failed upload	Idempotent request; existing successful record is recognised, not duplicated.
Two devices submit the same source record	Potential-duplicate flag raised; no automatic delete or overwrite.
Stale edit to an already-modified record	Version conflict; current server state is preserved and the record is routed for review.
QC/PI correction	Previous/new value and reason recorded in the audit log.
Locked record	No ordinary editing; a PI-authorised reopen requires a reason, and the action is logged.

11.2 Duplicate-detection strategy

Potential duplicates are detected using available combinations such as the encrypted/hashed hospital number, presentation date/time, age and sex. A match produces a review flag for an authorised reviewer — it never silently deletes or merges records, and restricted linkage information stays access-controlled throughout the review.

12. Security, Privacy & Data Governance

Control	Proposed implementation
Transport security	HTTPS/TLS for all mobile/web/API communication.
Authentication	Individual username/email + password; secure session/token management; optional PIN/biometric re-entry after initial login.
Authorisation	Role-based access control enforced on the Node.js server, not only in the app's interface.
Device management	One active registered device per RA account by default; an administrator can deactivate a lost/compromised device and authorise a replacement.
Local data	Encrypted local database and secure credential/token storage on-device.
Linkage privacy	Hospital-record identifiers are stored separately from analysis data and access-restricted.
Exports	No direct identifiers or raw linkage identifiers in research exports; RA accounts have no master-export capability.
Logging	Audit trail for record changes; technical sync logs for transmission events.
Third-party data leakage	No advertising SDKs or consumer analytics that transmit patient-derived information.
Backups	At least daily incremental and regular full backups, with a tested restore procedure.
Screens/notifications	Where feasible, block screenshots on linkage screens and suppress sensitive data in notifications.

13. Central Hosting & Repository Proposal

Proposed central repository: MongoDB hosted on an approved institutional or secure cloud environment. The Node.js API is the only application component permitted to communicate with the database.

Option	Description	Notes
A. Managed cloud (MongoDB Atlas)	3-node replica set, encryption at rest, automated continuous backup, selectable region	Fastest to set up; requires confirming the hosting region meets institutional data-governance policy.
B. Self-hosted institutional server	MongoDB replica set on LMUTH/university-approved infrastructure	Keeps data fully on institutional infrastructure; the institution provides and maintains the servers.

The final hosting provider and account arrangement require institutional approval. No identifiable research data will be stored in any personal developer or personal cloud account at any stage.

14. Analysis-Ready Export Structure

14.1 Primary deliverables

- LMUTH_SATS_MASTER_DATA.csv
- LMUTH_SATS_MASTER_DATA.xlsx
- LMUTH_SATS_Codebook.xlsx
- QC report
- Missing-data report
- Exclusion log
- Fieldwork progress report
- Optional SPSS .sav export, if approved and practical

14.2 Canonical export variable order

This is the study's own canonical one-row-per-presentation variable order from the approved Data Abstraction Tool:

```
study_id, ed_date, ed_time, triage_time, age, sex, referral, dm, htn, asthma, rvd,
other_comorb, comorb_any, preg_test, temp, hr, rr, sbp, dbp, spo2, rbs, rdt, mobility,
avpu, trauma, tews_total, complaint_text, complaint_group, multiple_complaint,
discriminator_yes, discriminator_type, sats_cat, initial_destination, clinician_time,
treatment_time, outcome24, outcome_datetime, outcome_source, outcome_verified,
sats_complete, miss_sats, miss_tews, miss_vitals, miss_outcome, source_conflict,
qc_verified, qc_reviewer
```

Export rule: one eligible ED presentation per row; no direct patient identifiers or raw linkage identifiers. Export filters will support all/completed/verified records, date range, SATS category, Research Assistant, pilot vs production, and eligible vs excluded. Note: complaint_text (verbatim chief complaint) is part of the tool's own canonical order — see Technical Question in §22 on whether this free text should be included in the PI-facing export as specified, given the small risk of incidental identifying detail in free text.

15. Quality Control & Independent Re-Abstraction

At least 10% independent QC is required — for approximately 8,000 eligible records, this is approximately 800 QC records. Selection should be random and, where possible, stratified by Research Assistant.

QC reviewers independently re-abstract key variables before seeing the RA's original values. The system automatically compares the two sets of entries, identifies discrepancies, and supports resolution. Key comparison fields: SATS category, TEWS, age, sex, referral, comorbidity, HR, RR, SBP, SpO2, discriminator, immediate destination and 24-hour outcome.

Outputs are generated in a form suitable for percentage agreement and Cohen's Kappa (categorical variables), and numeric difference/ICC-ready output (continuous variables).

16. Audit Trail & Data Provenance

Audit field	Requirement
Record ID	Study ID + internal UUID
Created by / on	User ID + timestamp
Device	Device ID
Modified by	User ID
Field changed	Variable name
Previous value	Retained
New value	Retained
Reason	Required for post-submission changes
App version	Version used at entry/edit

17. Pilot, Training & Production Modes

Mode	Purpose	Production data impact
TRAINING	Fictional/dummy cases for Research Assistant training.	Must never synchronise into production.
PILOT	15–20 record abstraction pilot to validate definitions, workflow and technical behaviour.	Automatically excluded from production exports.
PRODUCTION	Real study abstraction after definitions are approved/frozen.	Included according to export/QC status.

After field definitions are frozen, the production app version and data-dictionary version are recorded with every entry. Changes to variable meaning/coding after production begins require documented approval and an application-version increment.

18. Dashboards & Fieldwork Management

Dashboard	Proposed metrics / functions
Research Assistant	Assigned, completed, remaining, synced, pending sync, returned by QC, average completion time, records entered today/week.
PI	Total screened, eligible, excluded, abstracted, pending, synced, verified, locked; progress by RA; missingness; QC discrepancies; study profile; forecast.
QC	Assigned QC records, completion, discrepancy rates, agreement metrics, returned/resolved/verified counts.
Administrator	Users, devices, permissions, system configuration, technical audit and backup status.

19. Technical Validation Rules

- Hard validation rejects impossible entries such as SpO2 above 100% or a negative heart rate.
- Soft validation flags unusual-but-possible values and asks the collector to confirm against the source record — it never auto-corrects silently.
- Critical validation is repeated server-side during synchronisation, not trusted from the app alone.
- Malformed or logically inconsistent records are rejected with a clear correction message.
- The printed SATS 'Time to Treat' target is never treated as an actual treatment time; only documented actual timestamps are used.
- 'Missing from source record' is distinguishable from 'collector omission' using explicit Not Recorded/Missing states.

20. Acceptance & Multi-Device Test Plan

Scenario	Pass condition
Four-device offline creation	RA01–RA04 each create 10 records offline; after sync, the central repository contains exactly 40 unique records with correct attribution.
Interrupted sync	Network disabled mid-sync; local records remain safe; retry completes without duplicates.
Duplicate attempt	A likely duplicate is flagged; no silent overwrite.
Outcome pending	SATS/clinical data can be saved while the 24-hour outcome remains pending, then completed later.
QC blind re-entry	Reviewer independently abstracts key variables; comparison and discrepancy are logged.
Correction/audit	QC returns a record; RA corrects it; old/new values and reason remain in the audit trail.
Record locking	A verified record becomes locked; PI can reopen only with a reason, and the action is logged.
Device replacement	Admin deactivates a lost device and authorises a replacement without losing central data.
Export	PI downloads verified CSV/XLSX and codebook; one row per eligible presentation; no direct identifiers.

21. Implementation Phases

Phase	Deliverables
Phase 1 — Design sign-off	Architecture; schema/data dictionary; roles; screen flow; wireframes; offline/sync; duplicate strategy; security; hosting; sample exports; timeline/costing; technical questions.
Phase 2 — MVP	Login; roles; assignments; offline entry; abstraction screens; validation; local encryption; central sync; dashboard; CSV/XLSX export; audit trail.
Phase 3 — Pilot	At least four Android devices; 15–20 record pilot; simultaneous offline entry/sync; defect and definition correction.
Phase 4 — Production v1.0	Freeze dictionary/app; deployment; backups; training; live support.
Phase 5 — Analysis support	Final exports, codebook, missingness/QC reports and archival handover.

22. Technical Questions & Decisions Required

The items below are the remaining open decisions after reconciling this design against the approved Data Abstraction Tool. (That reconciliation already resolved several earlier questions — the complaint-group list, discriminator-type list, process-timestamp fields, comorbidity list and trauma coding are now fixed as shown in §5 and no longer listed here.)

#	Question / decision
1	Confirm which hospital identifier is permitted for restricted downstream outcome linkage.
2	Confirm the index-presentation/duplicate rule to apply when more than one potential record is identified for the same presentation.
3	Confirm source-priority rules when outcome evidence sources disagree (source_conflict = 1).
4	Confirm the approved institutional/secure cloud hosting environment (§13).
5	Confirm whether SPSS .sav export is required for MVP or remains optional/post-MVP.
6	Approve MongoDB as the proposed central database, given the source brief's own recommendation of PostgreSQL.
7	Confirm whether verbatim chief-complaint text (complaint_text) — included in the tool's own canonical export order — should appear in the PI-facing analysis export as specified, or be withheld/redacted given the small risk of incidental identifying detail in free text.
8	Confirm whether preg_test and rdt should be captured for every eligible record regardless of age/sex (using the tool's own 8 = Not done/not applicable code where irrelevant), or conditionally skipped in the app for records where they are clearly not

#	Question / decision
	applicable.

23. Assumptions & Scope Boundaries

- ✓ The approved Data Abstraction Tool is the authoritative source for all variable definitions, codes and abstraction rules; this design is built from it directly, and the codes reproduced in §5 are taken from it verbatim.
- ✓ Research decisions on variable meaning, coding, eligibility and outcome evidence will not be invented by the developer or coding agent.
- ✓ MongoDB is proposed as an approved equivalent central database technology; final institutional approval is required.
- ✓ The central repository is accessed through the Node.js API only.
- ✓ Pilot and training data remain segregated from production data.
- ✓ The analytical export is de-identified and contains one eligible ED presentation per row.
- ✓ The system is designed for at least 100,000 records and at least four concurrent Research Assistants, while allowing additional users.
- ✓ Costing and exact implementation duration will be finalised after approval of the complete data dictionary and hosting constraints.

24. Phase 1 Approval / Sign-Off Gate

The research team is requested to review and approve the Phase 1 design before complete MVP development begins. Approval should confirm that the system design matches the research questions, analytical model and approved Data Abstraction Tool.

Approval item	Approved / Comments
Proposed architecture	☐ Approved ☐ Changes required
MongoDB central repository	☐ Approved ☐ Changes required
Database schema / data dictionary	☐ Approved ☐ Changes required
User roles / permissions	☐ Approved ☐ Changes required
Mobile screen flow	☐ Approved ☐ Changes required
Wireframe specification	☐ Approved ☐ Changes required
Offline/synchronisation design	☐ Approved ☐ Changes required
Duplicate detection	☐ Approved ☐ Changes required
Security/privacy model	☐ Approved ☐ Changes required
Export structure/codebook approach	☐ Approved ☐ Changes required
Pilot/QC/acceptance approach	☐ Approved ☐ Changes required

Research Team / PI Signature

Name	Signature	Date

Decision: ☐ Approved ☐ Approved with changes ☐ Not approved

Comments:

Development gate: full MVP development will begin only after the research team has confirmed the data dictionary and screen flow, and approved this Phase 1 design.

Source basis: LMUTH SATS Research Data Platform | Developer Brief v1.0, including the specified research model, role model, offline/synchronisation requirements, data dictionary, QC, security, export, pilot, acceptance and development sign-off requirements.